

Blow-up Print: Rapidly 3D Printing Inflatable Objects in the Compressed State

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1 INTRODUCTION

3D printing is becoming a widely used technology not only for prototyping but for industrial fabrication. People have access to several types of 3D printing methods, such as fused deposition modeling (FDM), stereolithography apparatus (SLA), and selective laser sintering (SLS), to build physical objects with diverse material properties through different processes. Despite such options, we generally suffer from longer printing time, support material consumption, and/or storage space as a printed object gets bigger.

Thus, researchers have recently been working on achieving 3D objects while reducing these issues. For example, Yu et al. [2017] decomposed an input 3D object into a telescopic structure that can be used for better storage and transportation. Likewise, Pop-up Print [Noma et al. 2020] 3D printed objects in the compact folded state to reduce printing time and support material consumption. However, these methods are applicable only for the tapered region of the given 3D model and the resulting design space is limited.

On the other hand, pneumatically actuated inflatables have been explored to achieve deployable structures. For example, Panetta et al. [2021] achieved double-curved shape by inversely computing transformation of planar inflatable channels, which is fabricable by 2D fusing process. Also, Skouras et al. [2012] proposed the method to computationally design balloons inflatable into the target shapes.

Built upon these prior works, we propose Blow-up Print, the method to print a squashed 3D model and restore the original

shape by inflation in the real world. Squashing reduces height and volume, leading to a reduction in printing time and support material consumption. Compared to the previous methods [Noma et al. 2020; Yu et al. 2017], our proposing method is simple yet applicable to more general geometries other than tapered shapes.

2 OUR APPROACH

Our fabrication method consists of five steps: 1. squashing by elastic simulation, 2. solidification in voxel space, 3. unintentional contact reduction, 4. 3D printing, 5. inflation.

First, we vertically squash the input geometry in a simulation. We transform the input geometry into a surface mesh to reproduce the compression process by elastic simulation. A static disk is added to the lower part of the geometry to guide airflow. In the elastic simulation, we use a rigid rubber preset from Blender 3.0 and slowly move down a board to a predetermined height from directly above

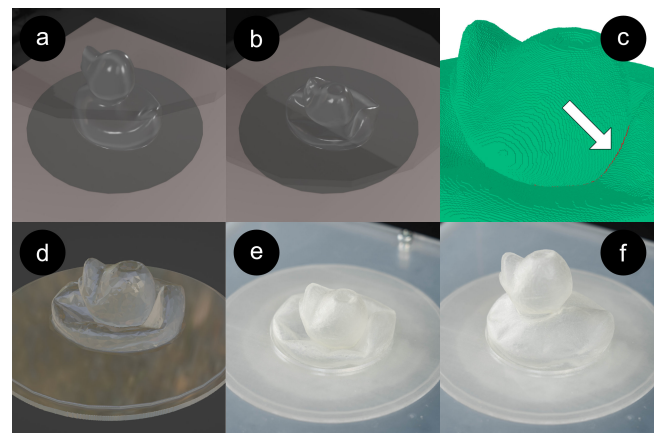


Figure 1: Our approach to fabricating an inflatable: (a-b) squashing the input with elastic simulation, (c) removing unintentional contacts in the voxel space (magnified view of the head), (d) converting the voxels back to the mesh, (e) 3D printing, and (f) inflating.

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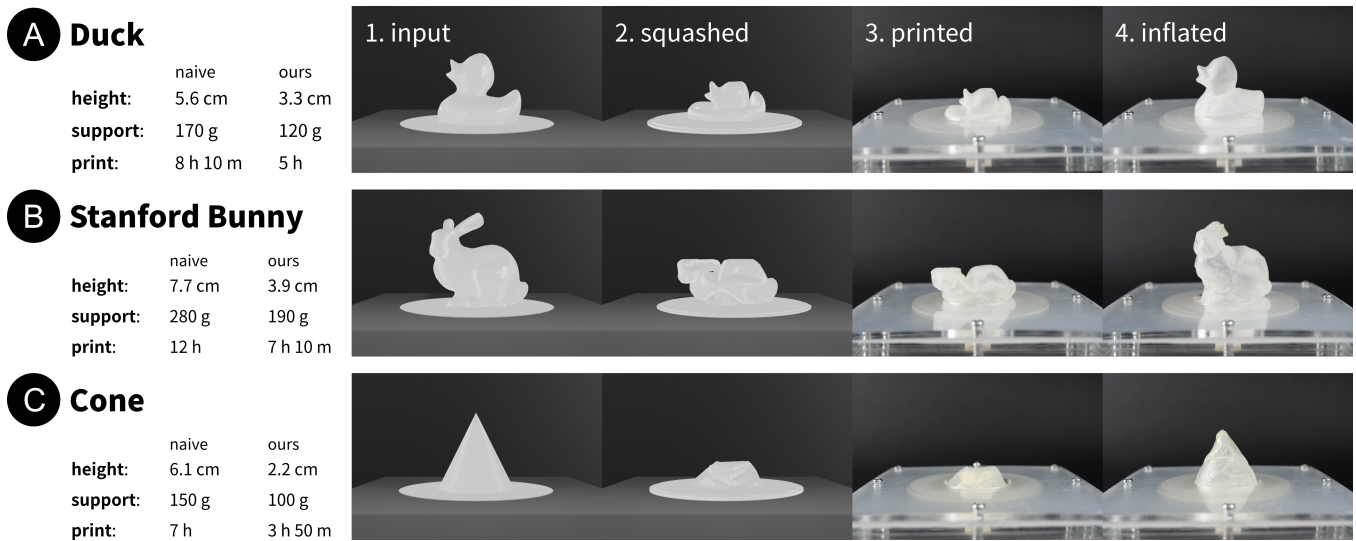


Figure 2: Examples of 3D-printed inflatables: (A) Duck, (B) Stanford Bunny, and (C) Cone. Each row includes the height of the object, the printing time and the amount of support material needed, and the images. The images follow the fabrication steps: (1) input model, (2) generated squashed model, (3) printed object, and (4) inflation result.

(Figure 1a-b). We set a weak gravity set vertically upward to keep the characteristics of the original shape as much as possible.

Next, the squashed mesh is placed in a $600 \times 600 \times 600$ 3D grid. Thickening is achieved by filling voxels whose distance to the compressed shape surface is less than a certain threshold. If this mesh is printed as is, the thickening may cause multiple surfaces to connect together and not expand as desired. To avoid such unintentional contact, an extra process is added to remove voxels that are far apart in on-surface path distance but adjacent to each other. In Figure 1c, voxels with unintentional contact are highlighted in red.

The voxels are converted back into a mesh by marching cubes (Figure 1d) and then printed using KEYENCE Agilista 3200 (Figure 1e). Soft silicone AR-G1L is used for the model body, and water-soluble resin AR-S1 is used as the support material. The reduction in height and volume leads to a reduction in printing time and support material consumption.

Finally, the printed object is set on a base frame to blow air inside and form the desired shape (Figure 1f). For simplicity, an airway is formed by sandwiching the printed material between two plastic panels, but it can alternatively be sealed like a balloon.

3 RESULTS

Our goals are to form a shape similar to the input with less time and support material consumption. We examined these with the three inputs shown in Figure 2. These models are printed at 12 cm in diameter and 1 mm in thickness and are compressed in height by 40%–60% compared to the originals. In all the examples, our method achieved similar shapes in less time and with less support material.

In the duck example, the occiput is pushed into the body during the compression step. The unintentional contact reduction algorithm eliminates these contacts and helps the duck inflate as expected. On the other hand, the skin is thinner there and is more likely to be breached when exposed to high air pressure.

In addition, some parts, such as bunny ears and straight parts of cones, have relatively low reproducibility. Possible reasons for this are that the elastic simulation does not consider the thickness of the material, the elasticity of silicone material is not taken into account, and some tiny unintentional contacts remain even after unintentional contact reduction.

4 CONCLUSION AND FUTURE WORK

The results shown in Figure 2 demonstrate that our method represents a significant step toward less time, less support material, and less storage space 3D printing by inflating compressed shapes.

However, there are still some issues with reproducibility and holes. These problems could be solved by developing elastic simulation methods to avoid self-collisions and structural vulnerabilities, or by improving the unintentional contact reduction algorithm.

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